LLM Research Training Program V2 (Strict Protocol Edition)

Core Philosophy

You are not learning to use models. You are learning to understand, analyze, and improve them. Every step requires hypothesis formation, experimentation, and mechanistic explanation.

Global Rules

1. You must attempt every concept before looking it up
2. You must explain your reasoning before asking questions
3. You may only use AI for boilerplate or debugging, not core logic
4. After using external help, you must re-implement from memory
5. Every experiment must include: hypothesis, result, interpretation

Assignment 1 — Minimal Self-Attention

Build a single-head self-attention model to solve argmax broadcasting.

1. Input: [2, 5, 1, 3] → Output: [5, 5, 5, 5]
2. Generalize to variable-length sequences
3. Embedding dimension ≤ 16
4. No LayerNorm, no residuals, no batching
5. Loss: MSE

Implementation Requirements

1. Manually implement Q, K, V projections
2. Compute attention scores QK^T
3. Apply scaling and softmax
4. Compute weighted sum with V

5 Train on synthetic dataset

Instrumentation Requirements

1 Log attention matrices during training
2 Track gradient norms of all weight matrices
3 Monitor output behavior (collapse, divergence, stability)

Failure Analysis (Required)

1 Describe what failed
2 Explain why you think it failed
3 Test at least two fixes
4 Evaluate which fix worked and why

Phase Progression

1 Phase 1: Single-head attention
2 Phase 2: Compare with non-attention models
3 Phase 3: Study optimization dynamics
4 Phase 4: Modify loss functions
5 Phase 5: Alter architecture
6 Phase 6: Design new training techniques

Mathematical Integration (V2)

1 Track gradient norms across layers
2 Analyze matrix products and spectral norms
3 Study loss surface curvature
4 Relate experiments to theory

Research Mindset

1. Always form hypotheses before experiments
2. Always question results
3. Always test assumptions
4. Document everything like a research paper

Final Objective

You should be able to independently analyze neural architectures, diagnose training failures, and propose novel improvements grounded in both theory and empirical evidence.